

Bare interaction of mono-layer graphene p_z orbital

I. BARE INTERACTION: MONOLAYER GRAPHENE p_z ORBITALS

We evaluate the bare Coulomb matrix elements

$$U_{abcd} = \int d\mathbf{r} d\mathbf{r}' \phi_a^*(\mathbf{r}) \phi_b(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \phi_c^*(\mathbf{r}') \phi_d(\mathbf{r}') \quad (1)$$

on the carbon p_z Wannier orbitals from Wannier90 ($5 \times 5 \times 1$ supercell, $150 \times 150 \times 192$ grid).

Reference values used for comparison:

- CoQui cRPA bare channels (Malte Rösner, k -mesh $16 \times 16 \times 1$, $n_b=144$, $c=15$).
- Wehling *et al.*, Phys. Rev. Lett. **106**, 236805 (2011), Table I (freestanding graphene, equilibrium lattice).

TABLE I: Bare p_z - p_z Coulomb matrix elements for monolayer graphene, in eV. “This work” uses `source_tol = volume_tol =  $10^{-3}$` on the raw Wannier90 XSF grid. Relative differences are quoted against the CoQui value.

Channel	Symbol	This work	CoQui (cRPA bare)	PRL 106, 236805	Δ vs. CoQui
On-site	U_{00} (onsite)	16.400	17.434	17.0	−5.93%
Nearest neighbor	U_{01} (NN)	8.529	8.840	8.5	−3.51%
Hund’s exchange (s.f.)	J^{sf}	0.1175	0.1308	–	−10.17%
Hund’s pair-hopping	J^{ph}	0.1175	0.1308	–	−10.17%

The two Hund’s channels agree numerically ($|J^{\text{sf}} - J^{\text{ph}}| < 10^{-16}$), confirming the real-orbital identity $V_{abba} = V_{abab}$. The difference with CoQui is likely because the CoQui cRPA run was under a periodic setup, which may have led to an overestimate of the bare interaction due to the periodic images. Only the on-site and NN channels are reported here— U_{02} , U_{03} require shifting the second orbital onto further-neighbor sublattice sites, which has not been done in this run.

II. TOLERANCE CONVERGENCE

The TKM3D quadrature tolerances `source_tol` and `volume_tol` were swept jointly from 10^{-3} to 10^{-5} .

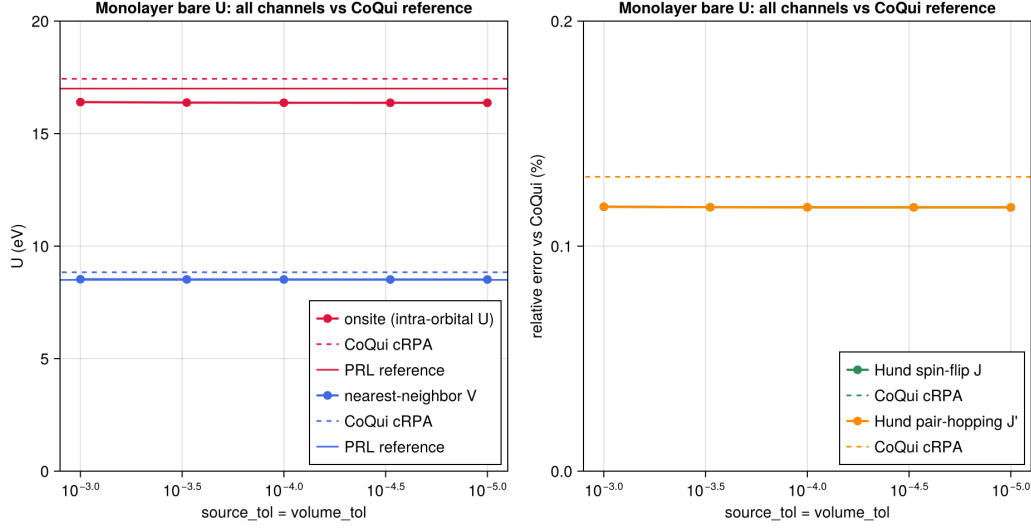


FIG. 1: All four bare channels (onsite, NN, Hund's spin-flip, Hund's pair-hopping) versus quadrature tolerance. Dots are this work; dashed horizontal lines are the CoQui cRPA bare references; and solid lines are the PRL 106, 236805 values.

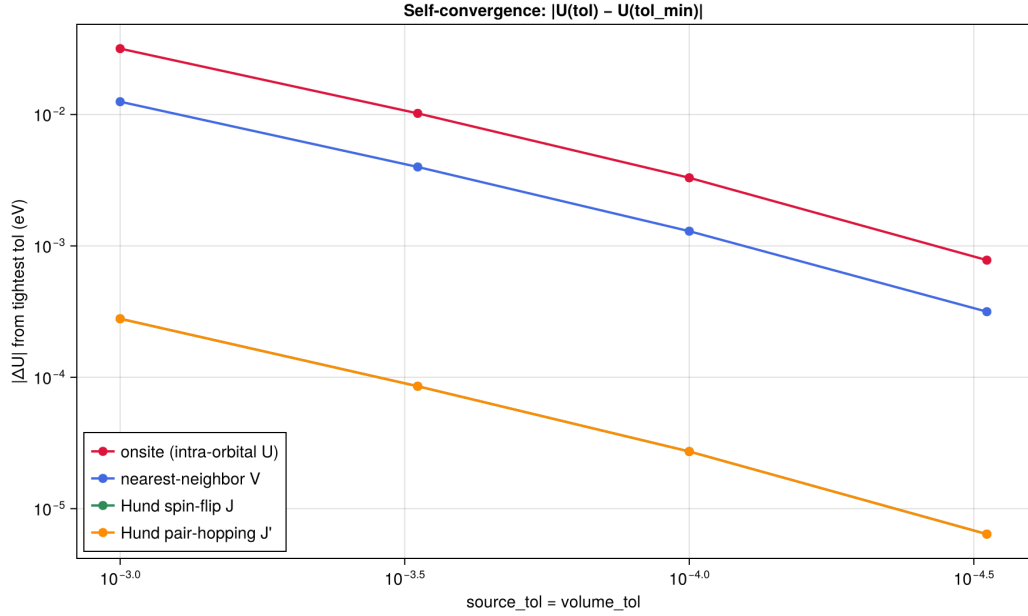


FIG. 2: Self-convergence of each channel: relative change of U between successive tolerance levels. A clear convergence is observed, so the discrepancy with CoQui in Table 1 is not a quadrature error.